Solution

Now, $(B \cap C) = \left\{\frac{1}{4}, 2, \frac{5}{2}\right\}$

$A \cap (B \cap C) = \left\{\frac{1}{4}, 2\right\}$ ... (1)

Then, $A \cap B = \left\{0, \frac{1}{4}, \frac{3}{4}, 2\right\}$

$(A \cap B) \cap C = \left\{\frac{1}{4}, 2\right\}$ ... (2)

From (1) and (2), it is verified that

$(A \cap B) \cap C = A \cap (B \cap C)$

Note

The set difference in general is not associative

that is, $(A-B)-C \neq A-(B-C)$.

But, if the sets A, B and C are mutually disjoint then the set difference is associative

that is, $(A-B)-C=A-(B-C)$.

Exercise 1.4

1. If $P = \{1,2,5,7,9\}$, $Q = \{2,3,5,9,11\}$, $R = \{3,4,5,7,9\}$ and $S = \{2,3,4,5,8\}$, then find

   (i) $(P \cup Q) \cup R$ (ii) $(P \cap Q) \cap S$ (iii) $(Q \cap S) \cap R$

2. Test for the commutative property of union and intersection of the sets
   $P = \{ x : x$ is a real number between 2 and 7$\}$ and
   $Q = \{ x : x$ is a rational number between 2 and 7$\}$.

3. If $A = \{p,q,r,s\}$, $B = \{m,n,q,s,t\}$ and $C = \{m,n,p,q,s\}$, then verify the associative property of union of sets.

4. Verify the associative property of intersection of sets for $A = \{-11, \sqrt{2}, \sqrt{5}, 7\}$, $B = \{\sqrt{3}, \sqrt{5}, 6, 13\}$ and $C = \{\sqrt{2}, \sqrt{3}, \sqrt{5}, 9\}$.

5. If $A=\{x : x = 2^n, n \in W$ and $n<4\}$, $B = \{x : x = 2n, n \in \mathbb{N}$ and $n \leq 4\}$ and

   $C = \{0,1,2,5,6\}$, then verify the associative property of intersection of sets.

1.6.3 Distributive Property

In lower classes, we have studied distributive property of multiplication over addition on numbers. That is, $a \times (b + c) = (a \times b) + (a \times c)$. In the same way we can define distributive properties on sets.

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